

第二节

定积分在几何学上的应用

- 一、平面图形的面积
- 二、平面曲线的弧长
- 三、已知平行截面面积函数的
立体体积
- 四、旋转体的侧面积 (补充)



一、平面图形的面积

1. 直角坐标情形

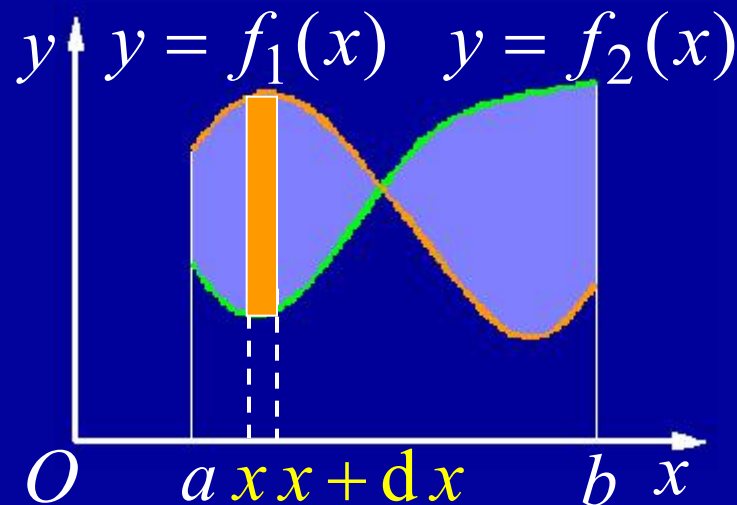
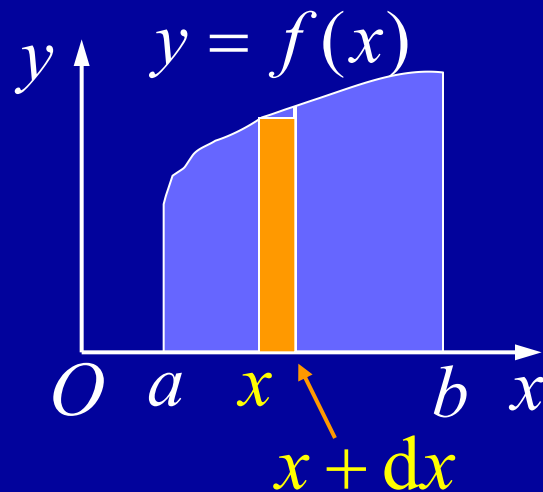
设曲线 $y = f(x) (\geq 0)$ 与直线 $x = a, x = b$ ($a < b$) 及 x 轴所围曲边梯形面积为 A , 则

$$dA = f(x) dx$$

$$A = \int_a^b f(x) dx$$

右下图所示图形面积为

$$A = \int_a^b |f_1(x) - f_2(x)| dx$$

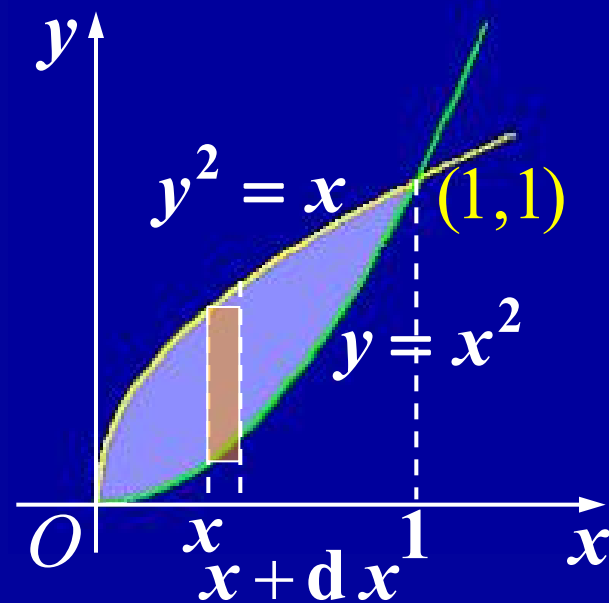


例1. 计算两条抛物线 $y^2 = x$, $y = x^2$ 在第一象限所围图形的面积.

解: 由 $\begin{cases} y^2 = x \\ y = x^2 \end{cases}$

得交点 $(0, 0)$, $(1, 1)$

$$\begin{aligned} \therefore A &= \int_0^1 (\sqrt{x} - x^2) dx \\ &= \left[\frac{2}{3} x^{\frac{3}{2}} - \frac{1}{3} x^3 \right]_0^1 \\ &= \frac{1}{3} \end{aligned}$$

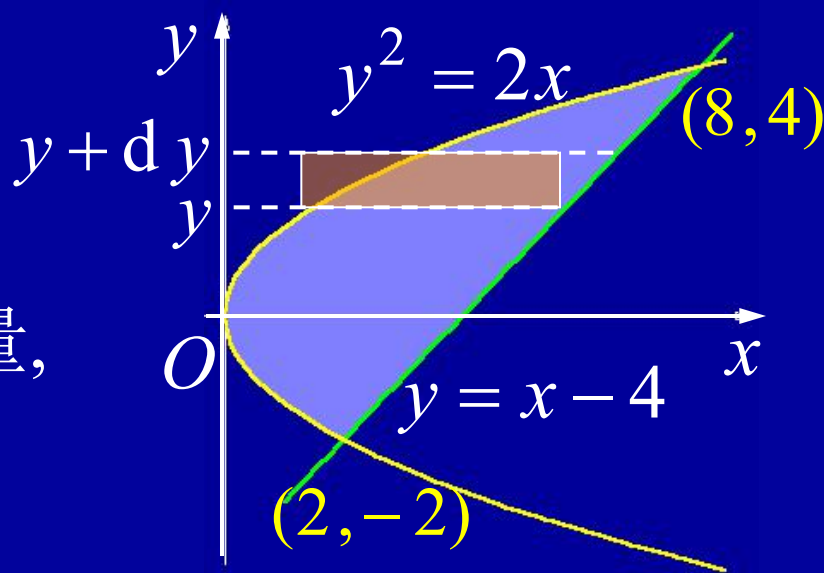


例2. 计算抛物线 $y^2 = 2x$ 与直线 $y = x - 4$ 所围图形的面积.

解: 由 $\begin{cases} y^2 = 2x \\ y = x - 4 \end{cases}$ 得交点
 $(2, -2), (8, 4)$

为简便计算, 选取 y 作积分变量,
则有

$$\begin{aligned} \therefore A &= \int_{-2}^4 \left(y + 4 - \frac{1}{2}y^2 \right) dy \\ &= \left[\frac{1}{2}y^2 + 4y - \frac{1}{6}y^3 \right]_{-2}^4 = 18 \end{aligned}$$



例3. 求椭圆 $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ 所围图形的面积.

解: 利用对称性, 有 $dA = y dx$

$$A = 4 \int_0^a y dx$$

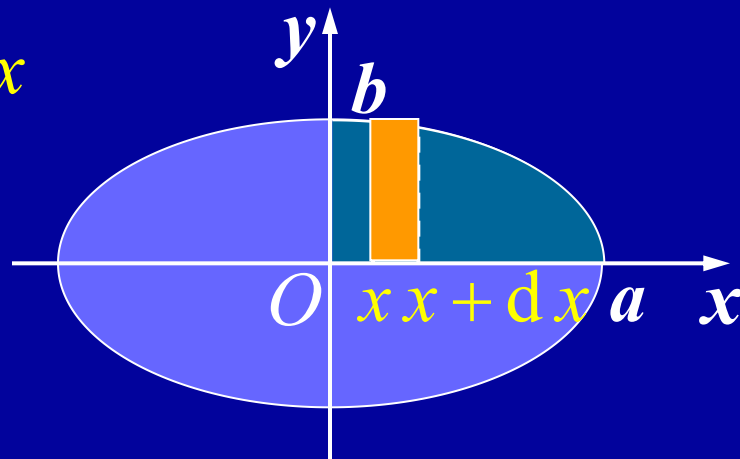
利用椭圆的参数方程

$$\begin{cases} x = a \cos t \\ y = b \sin t \end{cases} \quad (0 \leq t \leq 2\pi)$$

应用定积分换元法得

$$\begin{aligned} A &= 4 \int_{\frac{\pi}{2}}^0 b \sin t \cdot (-a \sin t) dt = 4ab \int_0^{\frac{\pi}{2}} \sin^2 t dt \\ &= 4ab \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \pi ab \end{aligned}$$

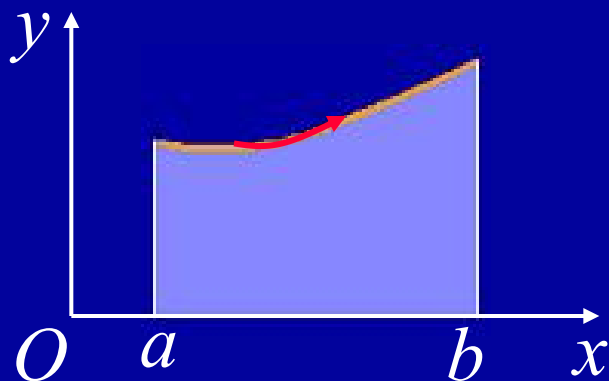
当 $a = b$ 时得圆面积公式



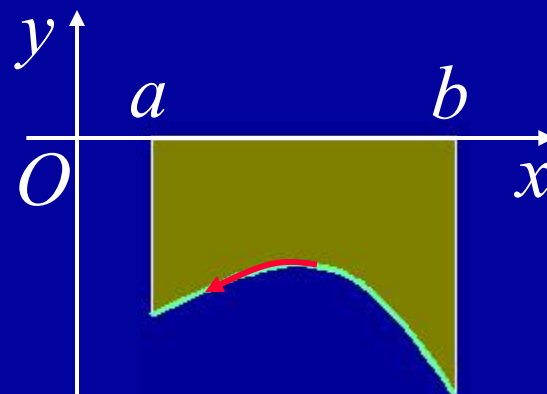
一般地, 当曲边梯形的曲边由参数方程

$$\begin{cases} x = \varphi(t) \\ y = \psi(t) \end{cases}$$

给出时, 按**顺时针方向**规定起点和终点的参数值 t_1, t_2



(t_1 对应 $x = a$)



(t_1 对应 $x = b$)

则曲边梯形面积 $A = \int_{t_1}^{t_2} \psi(t) \cdot \varphi'(t) dt$



例4. 求由摆线 $x = a(t - \sin t)$, $y = a(1 - \cos t)$ ($a > 0$) 的一拱与 x 轴所围平面图形的面积.

解: $A = \int_0^{2\pi} (1 - \cos t) \cdot a(1 - \cos t) dt$

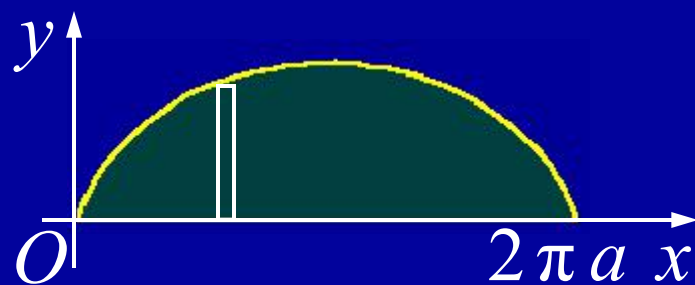
$$= a^2 \int_0^{2\pi} (1 - \cos t)^2 dt$$

$$= 4a^2 \int_0^{2\pi} \sin^4 \frac{t}{2} dt$$

$$= 8a^2 \int_0^{\pi} \sin^4 u du \quad (\text{令 } u = \frac{t}{2})$$

$$= 16a^2 \int_0^{\frac{\pi}{2}} \sin^4 u du$$

$$= 16a^2 \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} = 3\pi a^2$$



2. 极坐标情形

设 $\varphi(\theta) \in C[\alpha, \beta]$, $\varphi(\theta) \geq 0$, 求由曲线 $r = \varphi(\theta)$ 及射线 $\theta = \alpha, \theta = \beta$ 围成的曲边扇形的面积.

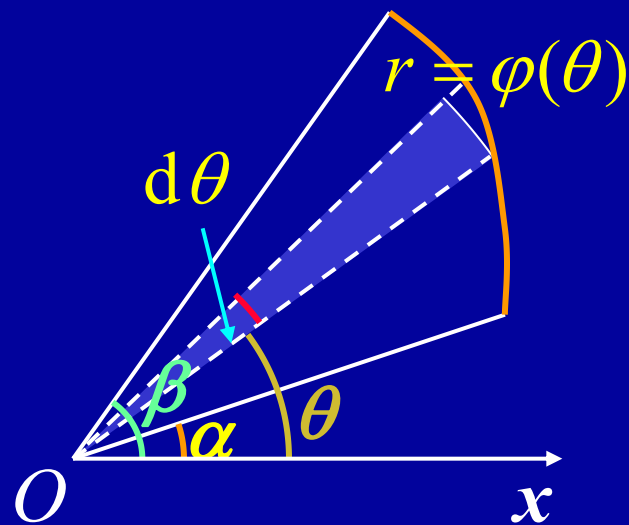
在区间 $[\alpha, \beta]$ 上任取小区间 $[\theta, \theta + d\theta]$

则对应该小区间上曲边扇形面积的近似值为

$$dA = \frac{1}{2} [\varphi(\theta)]^2 d\theta$$

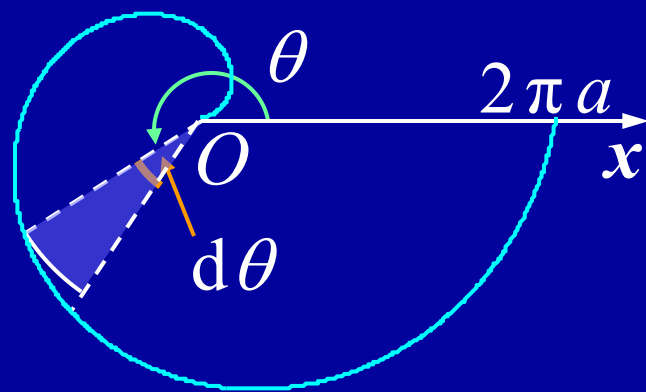
所求曲边扇形的面积为

$$A = \frac{1}{2} \int_{\alpha}^{\beta} \varphi^2(\theta) d\theta$$



例5. 计算阿基米德螺线 $r = a\theta$ ($a > 0$) 对应 θ 从 0 变到 2π 所围图形面积.

解:
$$A = \int_0^{2\pi} \frac{1}{2} (a\theta)^2 d\theta$$
$$= \frac{a^2}{2} \left[\frac{1}{3} \theta^3 \right]_0^{2\pi}$$
$$= \frac{4}{3} \pi^3 a^2$$



例6. 计算心形线 $r = a(1 + \cos \theta)$ ($a > 0$) 所围图形的面积. 心形线

解: $A = 2 \int_0^{\pi} \frac{1}{2} a^2 (1 + \cos \theta)^2 d\theta$

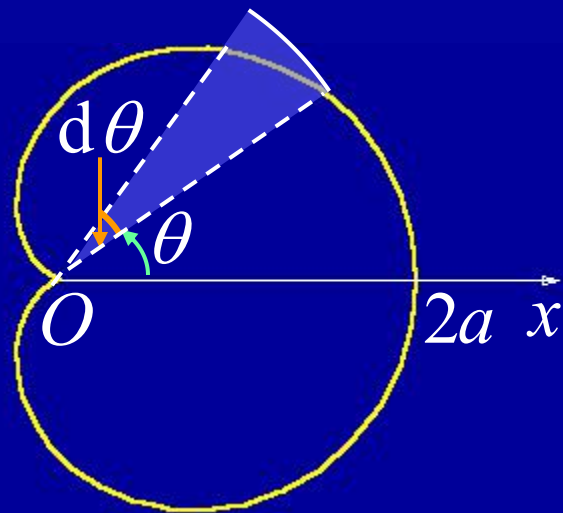
$$= a^2 \int_0^{\pi} 4 \cos^4 \frac{\theta}{2} d\theta$$

令 $t = \frac{\theta}{2}$

$$= 8a^2 \int_0^{\frac{\pi}{2}} \cos^4 t dt$$

$$= 8a^2 \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \frac{3}{2} \pi a^2$$

(利用对称性)

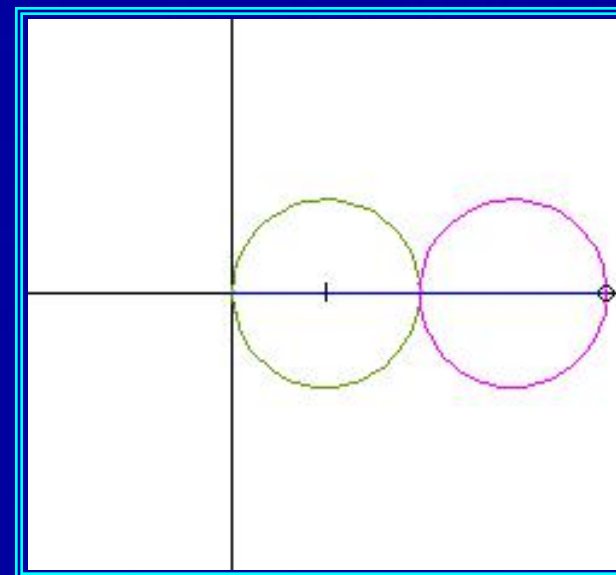
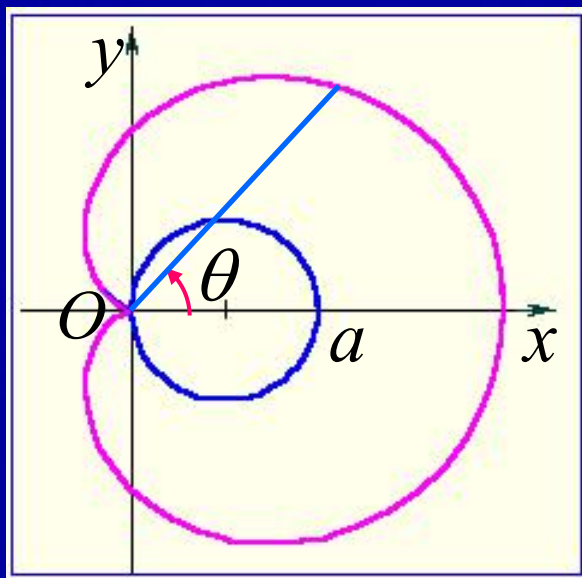


心形线(外摆线的一种)

$$x^2 + y^2 - ax = a\sqrt{x^2 + y^2}$$

$$\text{即 } r = a(1 + \cos \theta)$$

参数的几何意义



点击图中任意点
动画开始或暂停

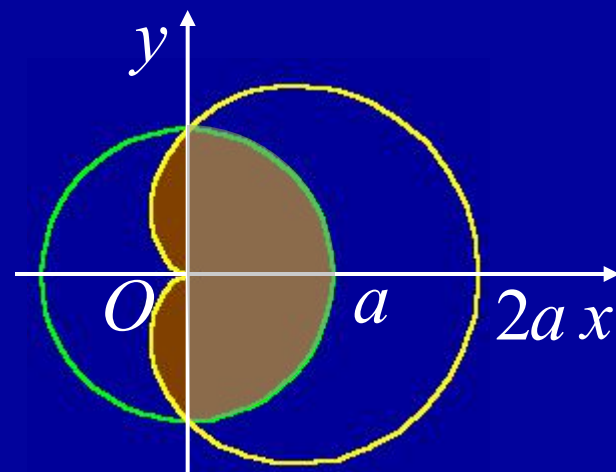
- 尖点: $(0,0)$
- 面积: $\frac{3}{2}\pi a^2$
- 弧长: $8a$



例7. 计算心形线 $r = a(1 + \cos \theta)$ ($a > 0$) 与圆 $r = a$ 所围图形的面积.

解: 利用对称性, 所求面积

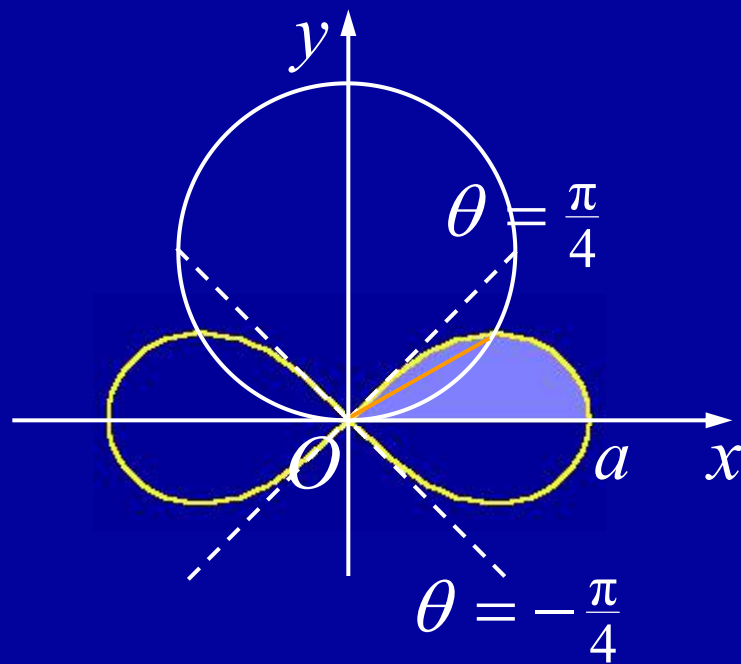
$$\begin{aligned} A &= \frac{1}{2} \pi a^2 + 2 \int_{\frac{\pi}{2}}^{\pi} \frac{1}{2} a^2 (1 + \cos \theta)^2 d\theta \\ &= \frac{1}{2} \pi a^2 + a^2 \int_{\frac{\pi}{2}}^{\pi} \left(\frac{3}{2} + 2 \cos \theta + \frac{1}{2} \cos 2\theta \right) d\theta \\ &= \frac{1}{2} \pi a^2 + a^2 \left(\frac{3}{4} \pi - 2 \right) \\ &= \frac{5}{4} \pi a^2 - 2a^2 \end{aligned}$$



例8. 求双纽线 $r^2 = a^2 \cos 2\theta$ 所围图形面积.

解: 利用对称性, 则所求面积为

$$\begin{aligned} A &= 4 \int_0^{\frac{\pi}{4}} \frac{1}{2} a^2 \cos 2\theta \, d\theta \\ &= a^2 \int_0^{\frac{\pi}{4}} \cos 2\theta \, d(2\theta) \\ &= a^2 [\sin 2\theta]_0^{\frac{\pi}{4}} = a^2 \end{aligned}$$



思考: 用定积分表示该双纽线与圆 $r = a\sqrt{2} \sin \theta$ 所围公共部分的面积.

答案: $A = 2 \left[\int_0^{\frac{\pi}{6}} a^2 \sin^2 \theta \, d\theta + \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{1}{2} a^2 \cos 2\theta \, d\theta \right]$



二、平面曲线的弧长

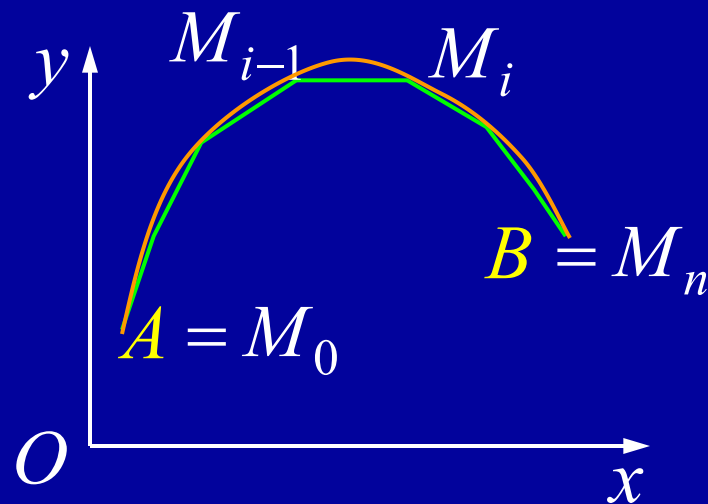
定义: 若在弧 $\widehat{AB}$ 上任意作内接折线, 当折线段的最大边长 $\lambda \rightarrow 0$ 时, 折线的长度趋向于一个确定的极限, 则称此极限为曲线弧 $\widehat{AB}$ 的弧长, 即

$$s = \lim_{\lambda \rightarrow 0} \sum_{i=1}^n |M_{i-1}M_i|$$

并称此曲线弧为可求长的.

定理: 任意光滑曲线弧都是可求长的.

(证明略)



(1) 曲线弧由直角坐标方程给出:

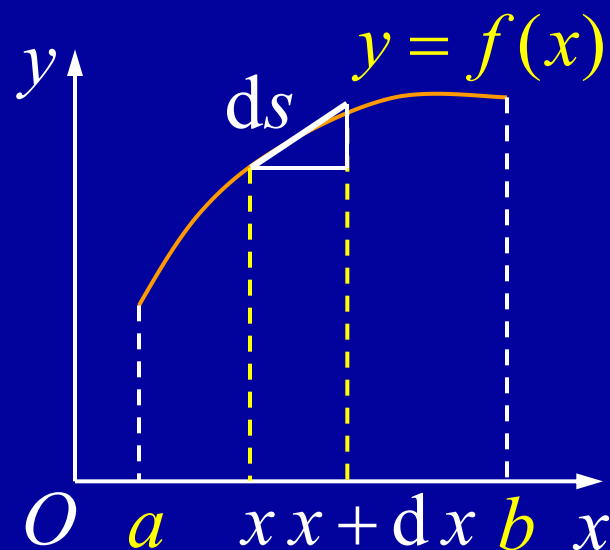
$$y = f(x) \quad (a \leq x \leq b)$$

弧长元素(弧微分):

$$\begin{aligned} ds &= \sqrt{(dx)^2 + (dy)^2} \\ &= \sqrt{1 + y'^2} dx \end{aligned}$$

因此所求弧长

$$\begin{aligned} s &= \int_a^b \sqrt{1 + y'^2} dx \\ &= \int_a^b \sqrt{1 + f'^2(x)} dx \end{aligned}$$



(2) 曲线弧由参数方程给出:

$$\begin{cases} x = \varphi(t) \\ y = \psi(t) \end{cases} \quad (\alpha \leq t \leq \beta)$$

弧长元素(弧微分):

$$\begin{aligned} ds &= \sqrt{(dx)^2 + (dy)^2} \\ &= \sqrt{\varphi'^2(t) + \psi'^2(t)} dt \end{aligned}$$

因此所求弧长

$$s = \int_{\alpha}^{\beta} \sqrt{\varphi'^2(t) + \psi'^2(t)} dt$$



(3) 曲线弧由极坐标方程给出:

$$r = r(\theta) \quad (\alpha \leq \theta \leq \beta)$$

令 $x = r(\theta)\cos\theta$, $y = r(\theta)\sin\theta$, 则得
弧长元素(弧微分):

$$\begin{aligned} ds &= \sqrt{[x'(\theta)]^2 + [y'(\theta)]^2} d\theta \\ &= \sqrt{r^2(\theta) + r'^2(\theta)} d\theta \quad (\text{自己验证}) \end{aligned}$$

因此所求弧长

$$s = \int_{\alpha}^{\beta} \sqrt{r^2(\theta) + r'^2(\theta)} d\theta$$



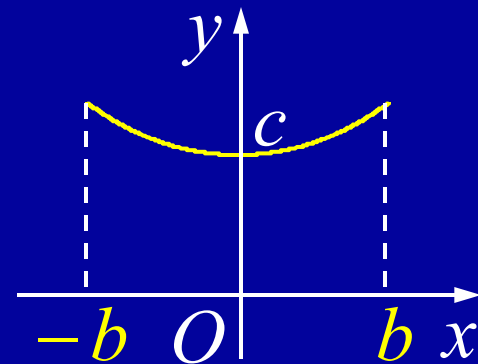
例9. 两根电线杆之间的电线, 由于其本身的重量, 下垂成悬链线. 悬链线方程为

$$y = c \operatorname{ch} \frac{x}{c} \quad (-b \leq x \leq b)$$

求这一段弧长.

解:

$$\begin{aligned} ds &= \sqrt{1 + y'^2} dx \\ &= \sqrt{1 + \operatorname{sh}^2 \frac{x}{c}} dx = \operatorname{ch} \frac{x}{c} dx \\ \therefore s &= 2 \int_0^b \operatorname{ch} \frac{x}{c} dx = 2c \left[\operatorname{sh} \frac{x}{c} \right]_0^b \\ &= 2c \operatorname{sh} \frac{b}{c} \end{aligned}$$



$$\begin{aligned} \operatorname{ch} x &= \frac{e^x + e^{-x}}{2} \\ \operatorname{sh} x &= \frac{e^x - e^{-x}}{2} \\ (\operatorname{ch} x)' &= \operatorname{sh} x \\ (\operatorname{sh} x)' &= \operatorname{ch} x \end{aligned}$$



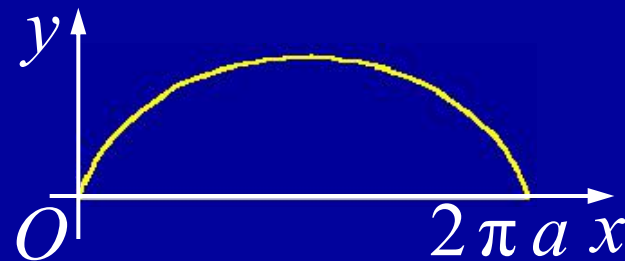
例10. 求连续曲线段 $y = \int_{-\frac{\pi}{2}}^x \sqrt{\cos t} \, dt$ 的弧长.

解: 此题 $\cos x \geq 0, \therefore -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

$$\begin{aligned} s &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{1 + y'^2} \, dx \\ &= 2 \int_0^{\frac{\pi}{2}} \sqrt{1 + (\sqrt{\cos x})^2} \, dx \\ &= 2 \int_0^{\frac{\pi}{2}} \sqrt{2} \cos \frac{x}{2} \, dx \\ &= 2\sqrt{2} \left[2 \sin \frac{x}{2} \right]_0^{\frac{\pi}{2}} \\ &= 4 \end{aligned}$$



例11. 计算摆线 $\begin{cases} x = a(t - \sin t) \\ y = a(1 - \cos t) \end{cases} (a > 0)$ 一拱 ($0 \leq t \leq 2\pi$) 的弧长.



解: $ds = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$

$$= \sqrt{a^2(1 - \cos t)^2 + a^2 \sin^2 t} dt$$

$$= a\sqrt{2(1 - \cos t)} dt$$

$$= 2a \sin \frac{t}{2} dt$$

$$\therefore s = \int_0^{2\pi} 2a \sin \frac{t}{2} dt = 2a \left[-2 \cos \frac{t}{2} \right]_0^{2\pi} = 8a$$



例12. 求阿基米德螺线 $r = a\theta$ ($a > 0$) 相应于 $0 \leq \theta \leq 2\pi$ 一段的弧长.

解: $ds = \sqrt{r^2(\theta) + r'^2(\theta)} d\theta$

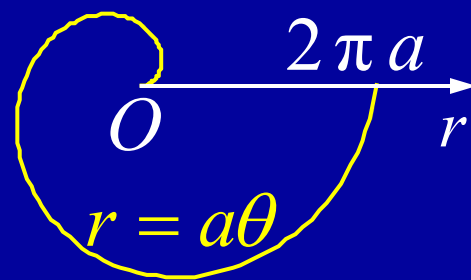
$$= \sqrt{a^2\theta^2 + a^2} d\theta$$

$$= a\sqrt{1 + \theta^2} d\theta$$

$$\therefore s = a \int_0^{2\pi} \sqrt{1 + \theta^2} d\theta$$

$$= a \left[\frac{\theta}{2} \sqrt{1 + \theta^2} + \frac{1}{2} \ln \left| \theta + \sqrt{1 + \theta^2} \right| \right]_0^{2\pi}$$

$$= a\pi\sqrt{1 + 4\pi^2} + \frac{a}{2} \ln(2\pi + \sqrt{1 + 4\pi^2})$$



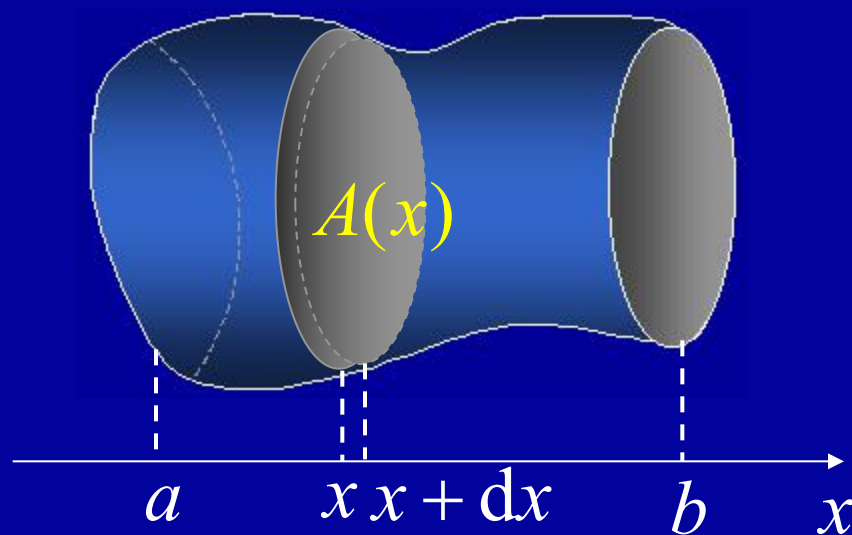
三、已知平行截面面积函数的立体体积

设所给立体垂直于 x 轴的截面面积为 $A(x)$, $A(x)$ 在 $[a, b]$ 上连续, 则对应于小区间 $[x, x + dx]$ 的体积元素为

$$dV = A(x) dx$$

因此所求立体体积为

$$V = \int_a^b A(x) dx$$



特别, 当考虑连续曲线段 $y = f(x)$ ($a \leq x \leq b$) 绕 x 轴旋转一周围成的立体体积时, 有

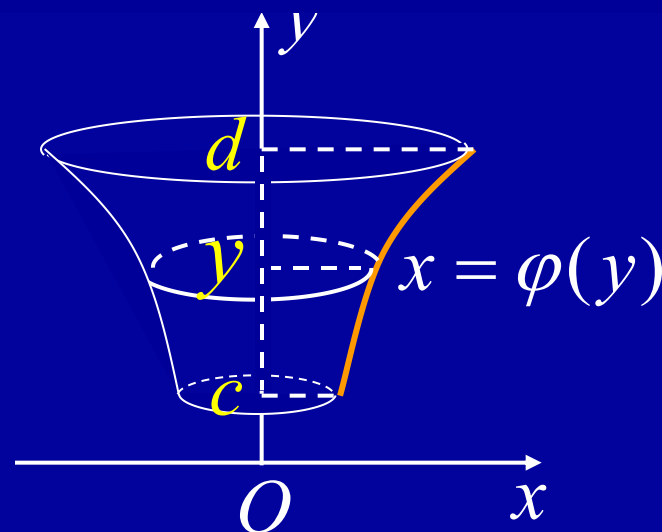
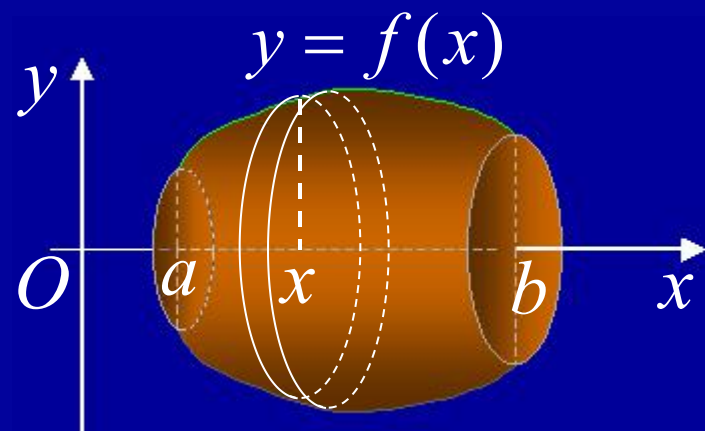
$$V = \int_a^b \pi [f(x)]^2 dx$$

当考虑连续曲线段

$$x = \varphi(y) \quad (c \leq y \leq d)$$

绕 y 轴旋转一周围成的立体体积时, 有

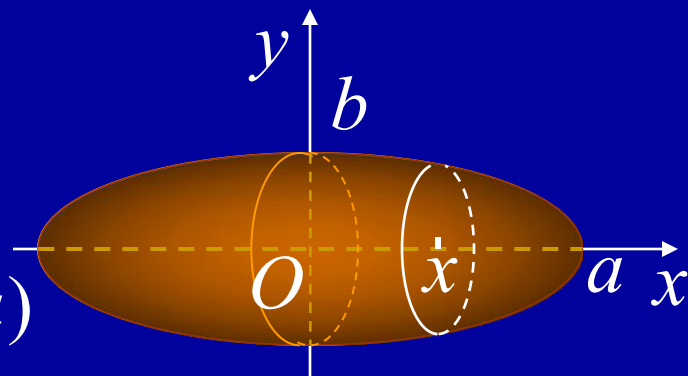
$$V = \int_c^d \pi [\varphi(y)]^2 dy$$



例13. 计算由椭圆 $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ 所围图形绕 x 轴旋转而转而成的椭球体的体积.

解: 方法1 利用直角坐标方程

$$y = \frac{b}{a} \sqrt{a^2 - x^2} \quad (-a \leq x \leq a)$$



则 $V = 2 \int_0^a \pi y^2 dx$ (利用对称性)

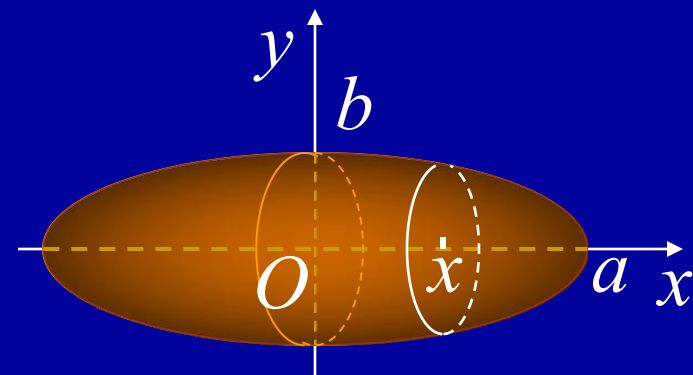
$$= 2\pi \frac{b^2}{a^2} \int_0^a (a^2 - x^2) dx$$

$$= 2\pi \frac{b^2}{a^2} \left[a^2 x - \frac{1}{3} x^3 \right]_0^a = \frac{4}{3} \pi a b^2$$



方法2 利用椭圆参数方程

$$\begin{cases} x = a \cos t \\ y = b \sin t \end{cases}$$



$$\begin{aligned} \text{则 } V &= 2 \int_0^a \pi y^2 dx = 2\pi \int_0^{\frac{\pi}{2}} ab^2 \sin^3 t dt \\ &= 2\pi ab^2 \cdot \frac{2}{3} \cdot 1 \\ &= \frac{4}{3} \pi ab^2 \end{aligned}$$

特别当 $b = a$ 时, 就得半径为 a 的球体的体积 $\frac{4}{3} \pi a^3$.

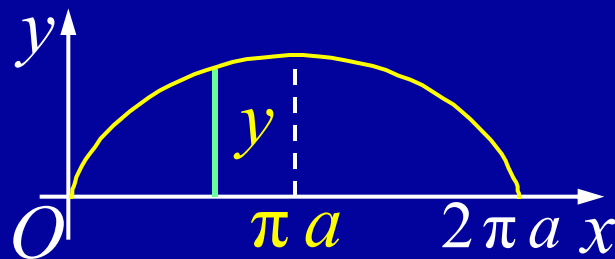


例14. 计算摆线 $\begin{cases} x = a(t - \sin t) \\ y = a(1 - \cos t) \end{cases} (a > 0)$ 的一拱与 $y=0$

所围成的图形分别绕 x 轴, y 轴旋转而成的立体体积.

解: 绕 x 轴旋转而成的体积为

$$V_x = \int_0^{2\pi a} \pi y^2 dx = 2 \int_0^{\pi a} \pi y^2 dx$$



利用对称性

$$= 2\pi \int_0^{\pi} a^2 (1 - \cos t)^2 \cdot a(1 - \cos t) dt$$

$$= 2\pi a^3 \int_0^{\pi} (1 - \cos t)^3 dt = 16\pi a^3 \int_0^{\pi} \sin^6 \frac{t}{2} dt \quad (\text{令 } u = \frac{t}{2})$$

$$= 32\pi a^3 \int_0^{\frac{\pi}{2}} \sin^6 u du = 32\pi a^3 \cdot \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2}$$

$$= 5\pi^2 a^3$$



$$\begin{cases} x = a(t - \sin t) \\ y = a(1 - \cos t) \end{cases} \quad (a > 0)$$

绕 y 轴旋转而成的体积为

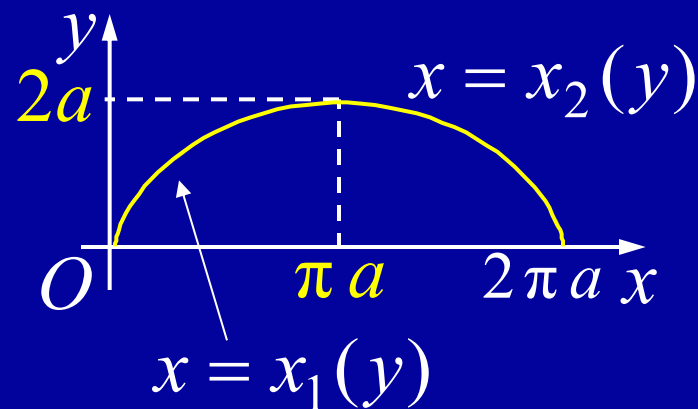
$$V_y = \int_0^{2a} \pi x_2^2(y) dy - \int_0^{2a} \pi x_1^2(y) dy$$

$$= \pi \int_{2\pi}^{\pi} a^2 (t - \sin t)^2 \cdot a \sin t dt$$

$$- \pi \int_0^{\pi} a^2 (t - \sin t)^2 \cdot a \sin t dt$$

$$= -\pi a^3 \int_0^{2\pi} (t - \sin t)^2 \sin t dt$$

$$= 6\pi^3 a^3 \quad \text{注}$$



注意上下限！

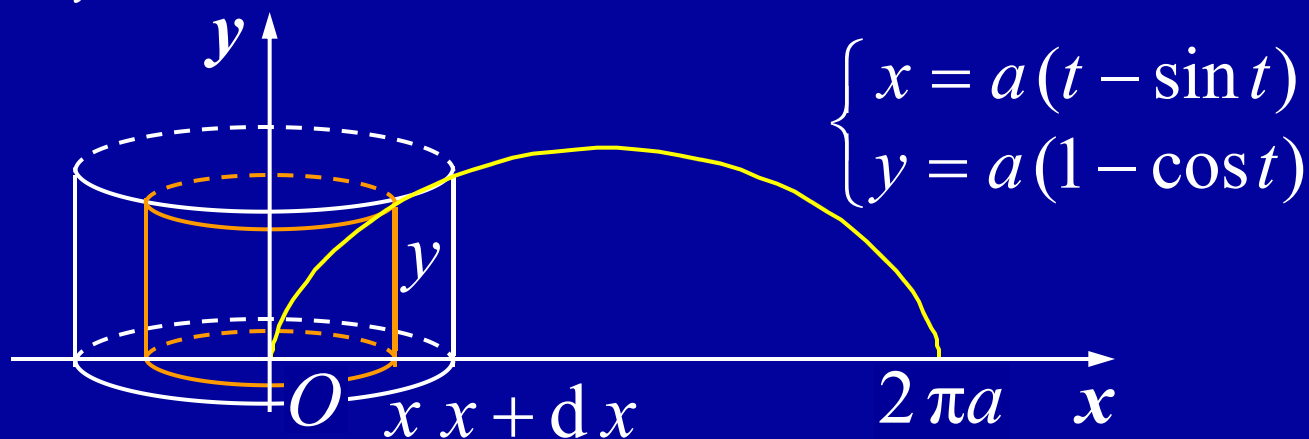


注

$$\begin{aligned}& \int_0^{2\pi} (t - \sin t)^2 \sin t \, dt \\&= \int_0^{2\pi} (t^2 \sin t - 2t \sin^2 t + \sin^3 t) \, dt \quad (\text{令 } u = t - \pi) \\&= \int_{-\pi}^{\pi} [-(\cancel{u^2} + 2\pi u + \cancel{\pi^2}) \sin u - 2(\cancel{u} + \pi) \sin^2 u - \cancel{\sin^3 u}] \, du \\&= -4\pi \underbrace{\int_0^{\pi} u \sin u \, du}_{\text{分部积分}} - 4\pi \underbrace{\int_0^{\pi} \sin^2 u \, du}_{\text{关于 } \frac{\pi}{2} \text{ 对称}} \quad (\text{利用 “偶倍奇零”}) \\&= -4\pi^2 - 8\pi \int_0^{\frac{\pi}{2}} \sin^2 u \, du \\&= -4\pi^2 - 8\pi \cdot \frac{1}{2} \cdot \frac{\pi}{2} = -6\pi^2\end{aligned}$$



说明: V_y 也可按柱壳法求出



柱面面积 $2\pi x \cdot y$

柱壳体积 $2\pi xy \cdot dx$

$$V_y = 2\pi \int_0^{2\pi a} xy dx$$

$$= 2\pi \int_0^{2\pi} a(t - \sin t) \cdot a^2(1 - \cos t)^2 dt$$



$$V_y =$$

$$= 2\pi \int_0^{2\pi} a(t - \sin t) \cdot a^2 (1 - \cos t)^2 dt$$

$$= 8\pi a^3 \int_0^{2\pi} (t - \sin t) \sin^4 \frac{t}{2} dt$$

$$\downarrow \text{令 } u = \frac{t}{2}$$

$$= 16\pi a^3 \int_0^{\pi} (2u - \sin 2u) \sin^4 u du$$

$$\downarrow \text{令 } v = u - \frac{\pi}{2}$$

$$= 16\pi a^3 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (\underbrace{2v}_{\text{奇函数}} + \underbrace{\pi}_{\text{奇函数}} + \underbrace{\sin 2v}_{\text{奇函数}}) \underbrace{\cos^4 v}_{\text{偶函数}} dv = 6\pi^3 a^3$$



例15. 设 $y = f(x)$ 在 $x \geq 0$ 时为连续的非负函数, 且 $f(0) = 0$, $V(t)$ 表示 $y = f(x)$, $x = t$ (> 0) 及 x 轴所围图形绕直线 $x = t$ 旋转一周所成旋转体体积, 证明:

$$V''(t) = 2\pi f(t).$$

证: 利用柱壳法

$$dV = 2\pi(t-x)f(x)dx$$

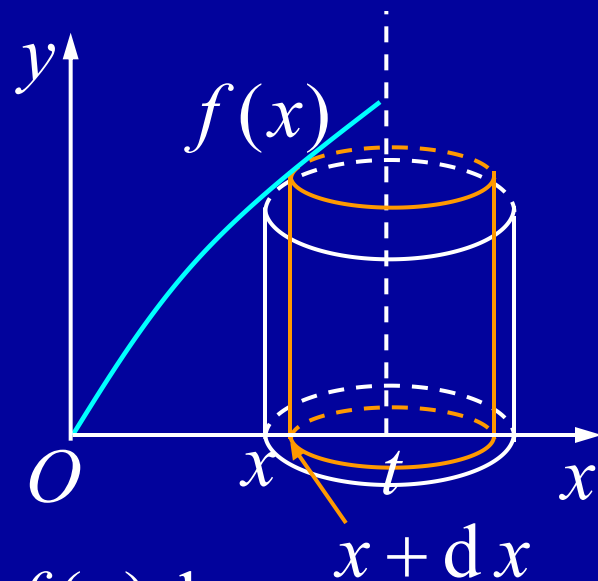
则

$$\begin{aligned} V(t) &= \int_0^t 2\pi(t-x)f(x)dx \\ &= 2\pi t \int_0^t f(x)dx - 2\pi \int_0^t x f(x)dx \end{aligned}$$

$$V'(t) = 2\pi \int_0^t f(x)dx + \cancel{2\pi t f(t)} - \cancel{2\pi t f(t)}$$

故

$$V''(t) = 2\pi f(t)$$



例16. 一平面经过半径为 R 的圆柱体的底圆中心, 并与底面交成 α 角, 计算该平面截圆柱体所得立体的体积.

解: 如图所示取坐标系, 则圆的方程为

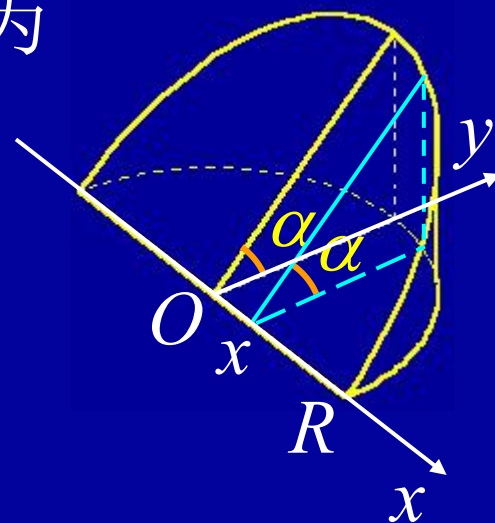
$$x^2 + y^2 = R^2$$

垂直于 x 轴的截面是直角三角形, 其面积为

$$A(x) = \frac{1}{2}(R^2 - x^2)\tan\alpha \quad (-R \leq x \leq R)$$

利用对称性

$$\begin{aligned} V &= 2 \int_0^R \frac{1}{2}(R^2 - x^2)\tan\alpha \, dx \\ &= 2 \tan\alpha \left[R^2 x - \frac{1}{3}x^3 \right]_0^R = \frac{2}{3}R^3 \tan\alpha \end{aligned}$$



思考: 可否选择 y 作积分变量?

此时截面面积函数是什么?

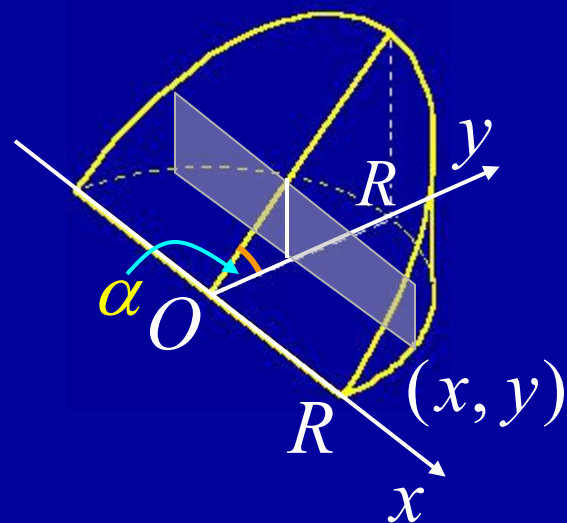
如何用定积分表示体积?

提示:

$$A(y) = 2x \cdot y \tan \alpha$$

$$= 2 \tan \alpha \cdot y \sqrt{R^2 - y^2}$$

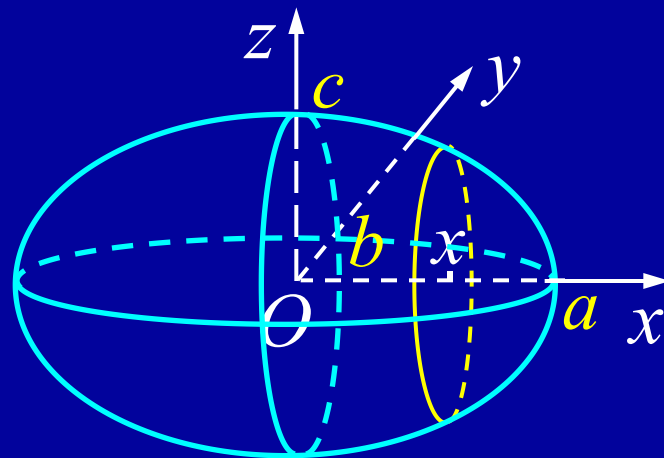
$$V = 2 \tan \alpha \cdot \int_0^R y \sqrt{R^2 - y^2} \, dy$$



例17. 计算由曲面 $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ 所围立体(椭球体)的体积.

解: 垂直 x 轴的截面是椭圆

$$\frac{y^2}{b^2(1-\frac{x^2}{a^2})} + \frac{z^2}{c^2(1-\frac{x^2}{a^2})} = 1$$



它的面积为 $A(x) = \pi bc(1 - \frac{x^2}{a^2})$ $(-a \leq x \leq a)$

因此椭球体体积为

$$V = 2 \int_0^a \pi bc(1 - \frac{x^2}{a^2}) dx = 2\pi bc \left[x - \frac{x^3}{3a^2} \right]_0^a = \frac{4}{3} \pi abc$$

特别当 $a = b = c$ 时就是球体体积.



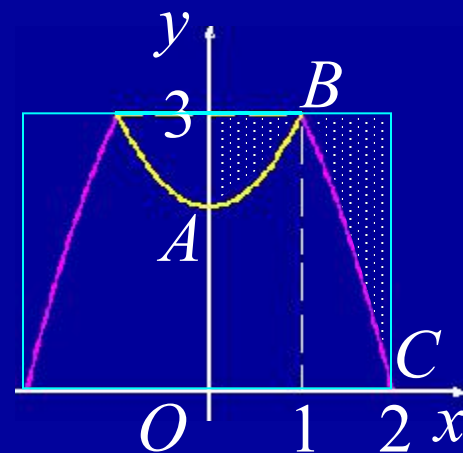
例18. 求曲线 $y = 3 - |x^2 - 1|$ 与 x 轴围成的封闭图形绕直线 $y=3$ 旋转得的旋转体体积. (1994 考研)

解: 利用对称性, 在第一象限

$$y = \begin{cases} x^2 + 2, & 0 \leq x \leq 1 \\ 4 - x^2, & 1 < x \leq 2 \end{cases}$$

故旋转体体积为

$$\begin{aligned} V &= \pi \cdot 3^2 \cdot 4 - 2 \int_0^1 \pi [3 - (x^2 + 2)]^2 dx \\ &\quad - 2 \int_1^2 \pi [3 - (4 - x^2)]^2 dx \\ &= 36\pi - 2\pi \int_0^2 (x^2 - 1)^2 dx = \frac{448}{15}\pi \end{aligned}$$



四、旋转体的侧面积 (补充)

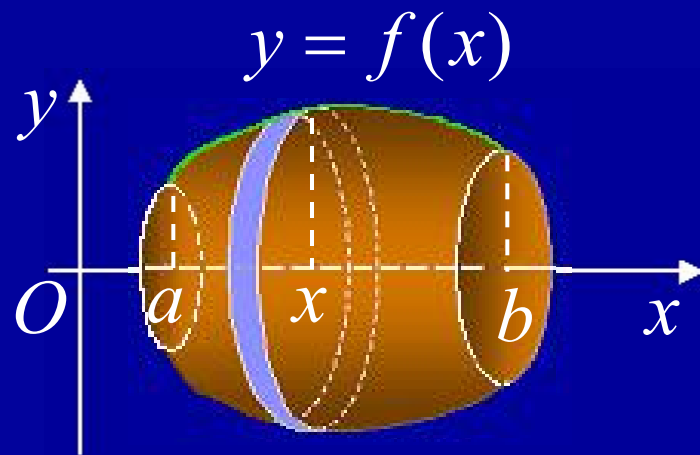
设平面光滑曲线 $y = f(x) \in C^1[a, b]$, 且 $f(x) \geq 0$, 求它绕 x 轴旋转一周所得到的旋转曲面的侧面积.

取侧面积元素: 位于 $[x, x + dx]$ 上的圆台的侧面积

$$\begin{aligned} dS &= 2\pi y ds \\ &= 2\pi f(x) \sqrt{1 + f'^2(x)} dx \end{aligned}$$

积分后得旋转体的侧面积

$$S = 2\pi \int_a^b f(x) \sqrt{1 + f'^2(x)} dx$$



注意：侧面积元素

$$dS = 2\pi y ds \neq 2\pi y dx$$

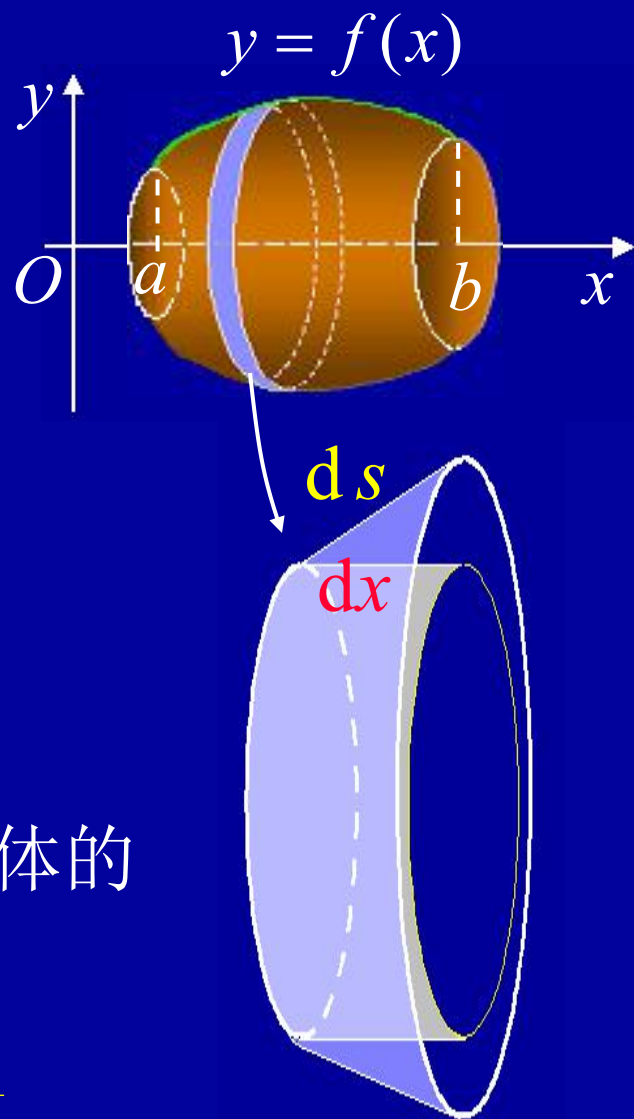
原因是 $2\pi y dx$ 不是薄片侧面积 ΔS 的线性主部.

若光滑曲线由参数方程

$$\begin{cases} x = \varphi(t) \\ y = \psi(t) \end{cases} \quad (\alpha \leq t \leq \beta)$$

给出, 则它绕 x 轴旋转一周所得旋转体的侧面积为

$$S = \int_{\alpha}^{\beta} 2\pi \psi(t) \sqrt{\varphi'^2(t) + \psi'^2(t)} dt$$



例19. 计算圆 $x^2 + y^2 = R^2$ 在 $x \in [x_1, x_2] \subset [-R, R]$ 上绕 x 轴旋转一周所得的球台的侧面积 S .

解: 对曲线弧

$$y = \sqrt{R^2 - x^2}, \quad x \in [x_1, x_2]$$

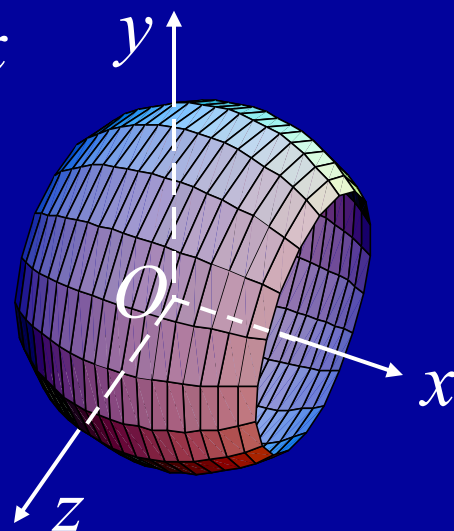
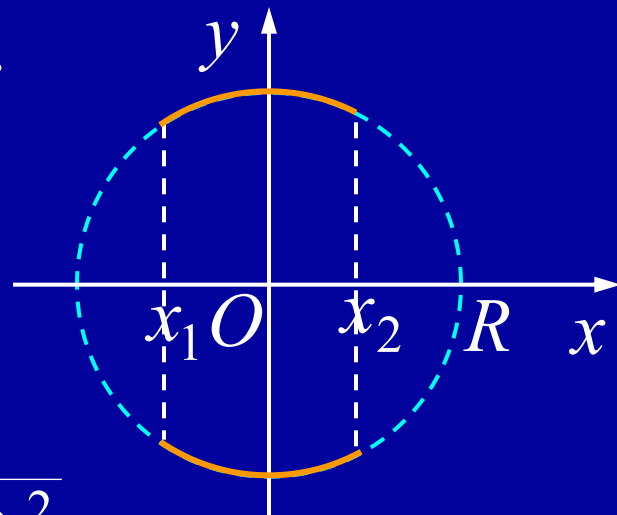
应用公式得

$$S = 2\pi \int_{x_1}^{x_2} \sqrt{R^2 - x^2} \cdot \sqrt{1 + \left(\frac{-x}{\sqrt{R^2 - x^2}} \right)^2} dx$$

$$= 2\pi \int_{x_1}^{x_2} R dx = 2\pi R(x_2 - x_1)$$

当球台高 $h = 2R$ 时, 得球的表面积公式

$$S = 4\pi R^2$$

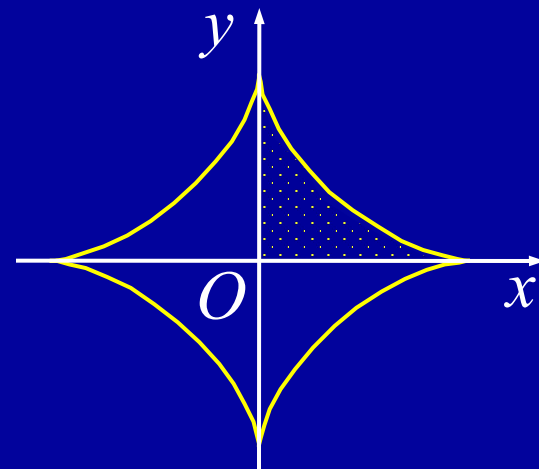


例20. 求由星形线 $x = a \cos^3 t$, $y = a \sin^3 t$ 绕 x 轴旋转一周所得的旋转体的表面积 S .

星形线

解: 利用对称性

$$\begin{aligned} S &= 2 \cdot 2\pi \int_0^{\frac{\pi}{2}} a \sin^3 t \\ &\quad \cdot \sqrt{(-3a \cos^2 t \sin t)^2 + (3a \sin^2 t \cos t)^2} dt \\ &= 12\pi a^2 \int_0^{\frac{\pi}{2}} \sin^4 t \cos t dt \\ &= 12\pi a^2 \left[\frac{1}{5} \sin^5 t \right]_0^{\frac{\pi}{2}} = \frac{12}{5} \pi a^2 \end{aligned}$$



$$\begin{cases} x = \varphi(t) \\ y = \psi(t) \end{cases} \quad (\alpha \leq t \leq \beta)$$

$$S = \int_{\alpha}^{\beta} 2\pi \psi(t) \sqrt{\varphi'^2(t) + \psi'^2(t)} dt$$



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星形线



目录



上页



下页



返回



结束

星形线 $x = a \cos^3 t, y = a \sin^3 t$

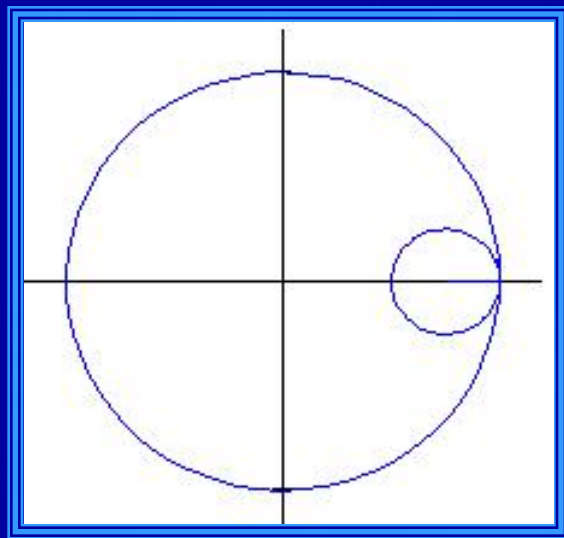
星形线是内摆线的一种.(当小圆在圆内沿圆周滚动时,小圆上的定点的轨迹为内摆线)

大圆半径

$$R = a$$

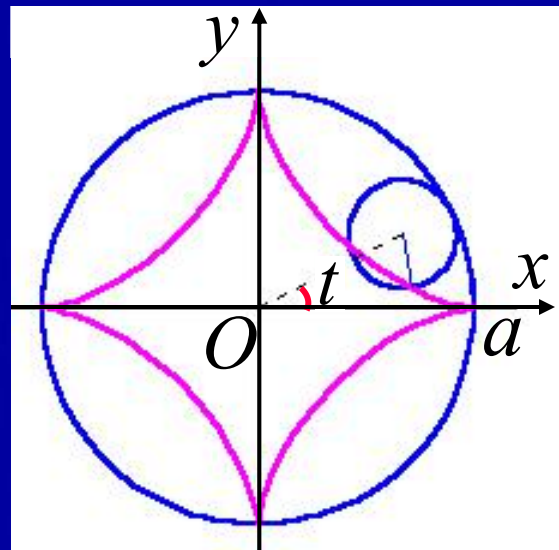
小圆半径

$$r = \frac{a}{4}$$



点击图片任意处
播放开始或暂停

参数的几何意义



内容小结

上下限按顺时针方向
确定

1. 平面图形的面积

边界方程

- 直角坐标方程
- 参数方程 $A = \int_{t_1}^{t_2} \psi(t) \cdot \varphi'(t) dt$
- 极坐标方程 $A = \frac{1}{2} \int_{\alpha}^{\beta} \varphi^2(\theta) d\theta$

2. 平面曲线的弧长

弧微分: $ds = \sqrt{(dx)^2 + (dy)^2}$

注意: 求弧长时积分上下限必须上大下小

曲线方程

- 直角坐标方程
- 参数方程
- 极坐标方程 $ds = \sqrt{r^2(\theta) + r'^2(\theta)} d\theta$



3. 已知平行截面面积函数 $A(x)$ 的立体体积

$$V = \int_a^b A(x) dx$$

—————> 旋转体的体积

$$y = y(x) \begin{cases} \text{绕 } x \text{ 轴: } A(x) = \pi y^2 \\ \text{绕 } y \text{ 轴: } A(x) = 2\pi xy \quad (\text{柱壳法}) \end{cases}$$

4. 旋转体的侧面积

$y = y(x)$ 绕 x 轴旋转, 侧面积元素为 $dS = 2\pi y ds$

(注意在不同坐标系下 ds 的表达式)



思考与练习

1. 用定积分表示图中阴影部分的面积 A 及边界长 s .

提示: 交点为 $(1, -1)$, $(9, 3)$, 以 x 为积分变量, 则要分两段积分, 故以 y 为积分变量.

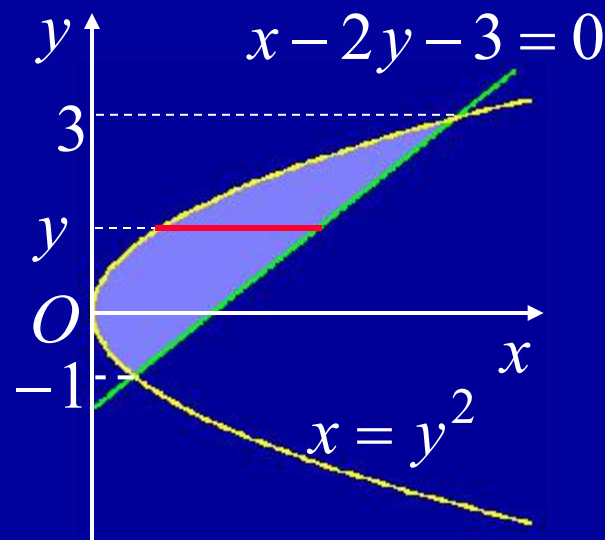
$$A = \int_{-1}^3 [(2y+3) - y^2] dy = \frac{32}{3}$$

弧线段部分

直线段部分

$$s = \int_{-1}^3 \sqrt{1+4y^2} dy + \int_{-1}^3 \sqrt{1+2^2} dy$$

$$= 3\sqrt{37} + 5\sqrt{5} + \frac{1}{4} [\ln(6 + \sqrt{37}) + \ln(2 + \sqrt{5})]$$



2. 试用定积分求圆 $x^2 + (y-b)^2 = R^2$ ($R < b$) 绕 x 轴旋转而成的环体体积 V 及表面积 S .

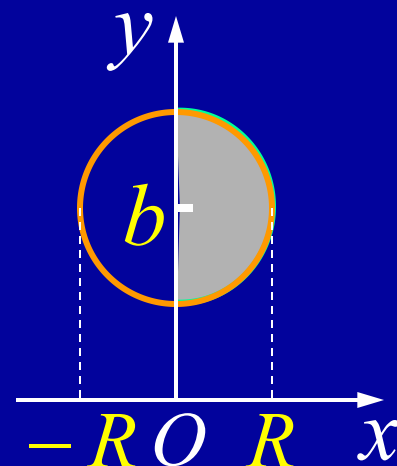
提示: $\frac{\text{上}}{\text{下}}$ 半圆为 $y = b \pm \sqrt{R^2 - x^2}$

$$y' = \mp \frac{x}{\sqrt{R^2 - x^2}}$$

求体积:

方法1 利用对称性

$$\begin{aligned} V &= 2 \int_0^R \pi \left[(b + \sqrt{R^2 - x^2})^2 - (b - \sqrt{R^2 - x^2})^2 \right] dx \\ &= 2\pi^2 R^2 b \end{aligned}$$



上
下 半圆为 $y = b \pm \sqrt{R^2 - x^2}$, $y' = \mp \frac{x}{\sqrt{R^2 - x^2}}$

方法2 用柱壳法

$$dV = 2\pi y \cdot 2x \cdot dy$$

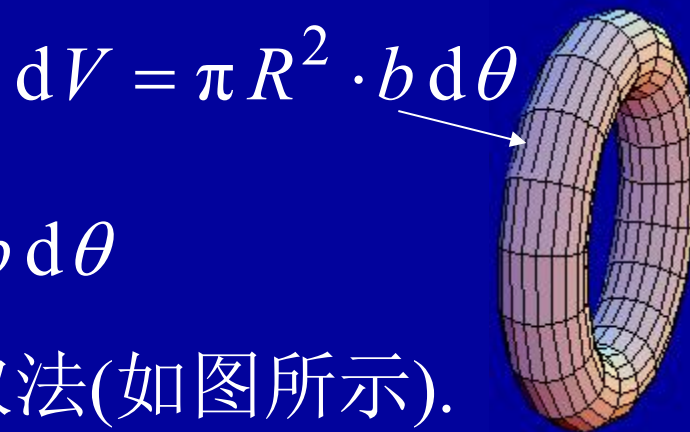
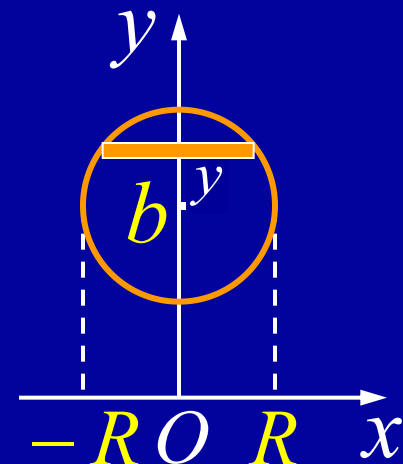
$$V = 4\pi \int_{b-R}^{b+R} y \sqrt{R^2 - (y-b)^2} dy$$

$$= 2\pi^2 R^2 b$$

说明: 上式可变形为

$$V = \pi R^2 \cdot 2\pi b = \int_0^{2\pi} \pi R^2 \cdot b d\theta$$

此式反映了环体元素的另一种取法(如图所示).



上
下 半圆为 $y = b \pm \sqrt{R^2 - x^2}$, $y' = \mp \frac{x}{\sqrt{R^2 - x^2}}$

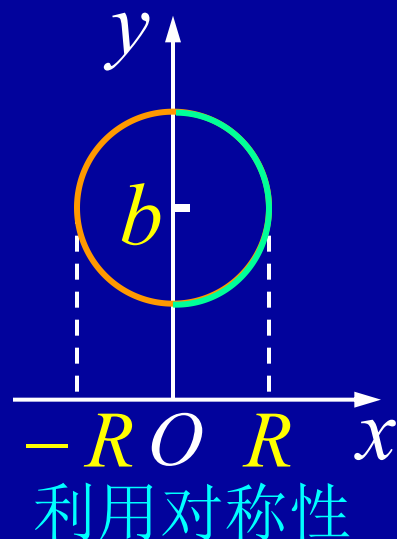
求侧面积：

$$S = 2 \int_0^R 2\pi (b + \sqrt{R^2 - x^2}) \cdot \sqrt{1 + y'^2} dx$$

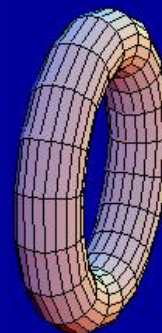
$$+ 2 \int_0^R 2\pi (b - \sqrt{R^2 - x^2}) \cdot \sqrt{1 + y'^2} dx$$

二者 y'^2 相同

$$= 8\pi b \int_0^R \sqrt{1 + y'^2} dx = 4\pi^2 bR$$



上式也可写成 $S = 2\pi R \cdot 2\pi b = \int_0^{2\pi} 2\pi R \cdot b d\theta$



它也反映了环面元素的另一种取法: $dS = 2\pi R \cdot b d\theta$



作业

面积及弧长部分:

P284 2 (1), (3); 3; 4; 5 (2), (3); 8 (2);
9; 10; 22; 25; 27; 30

体积及表面积部分:

P286 13; 14; 15 (1), (4); 17; 18

补充题: 设有曲线 $y = \sqrt{x-1}$, 过原点作其切线, 求由此曲线、切线及 x 轴围成的平面图形绕 x 轴旋转一周所得到的旋转体的表面积.



备用题 1. 求曲线 $|\ln x| + |\ln y| = 1$ 所围图形的面积.

解: 显然 $|\ln x| \leq 1, |\ln y| \leq 1$

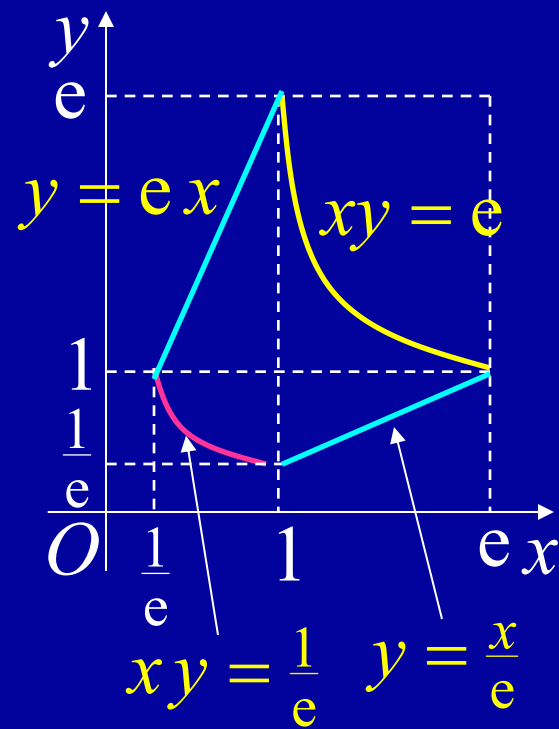
$$\Rightarrow e^{-1} \leq x \leq e, e^{-1} \leq y \leq e$$

$$\text{又 } |\ln x| = \begin{cases} \ln x, & 1 \leq x \leq e \\ -\ln x, & e^{-1} \leq x \leq 1 \end{cases}$$

$$|\ln y| = \begin{cases} \ln y, & 1 \leq y \leq e \\ -\ln y, & e^{-1} \leq y \leq 1 \end{cases}$$

故在区域 $\begin{cases} e^{-1} \leq x \leq 1 \\ e^{-1} \leq y \leq 1 \end{cases}$ 中曲线为 $xy = \frac{1}{e}$, 同理其他.

面积为
$$S = \int_{\frac{1}{e}}^1 (ex - \frac{1}{ex}) dx + \int_1^e (\frac{e}{x} - \frac{x}{e}) dx = e - \frac{1}{2e} - \frac{1}{2}$$



2. λ 为何值才能使 $y = x(x-1)$ 与 x 轴围成的面积等于 $y = x(x-1)$ 与 $x = \lambda$ 及 x 轴围成的面积.

解: $y = x(x-1)$ 与 x 轴所围面积

$$A_1 = \int_0^1 -x(x-1) dx = \frac{1}{6}$$

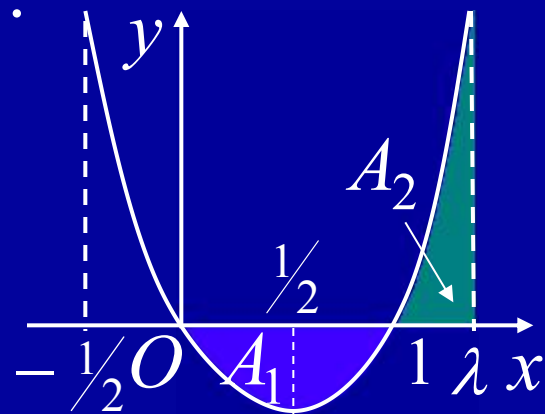
$\lambda \geq 0$ 时,

$$A_2 = \int_1^\lambda x(x-1) dx = \frac{1}{3}\lambda^3 - \frac{1}{2}\lambda^2 + \frac{1}{6}$$

由 $A_1 = A_2$, 得 $\lambda^2(\frac{1}{3}\lambda - \frac{1}{2}) = 0$, 故

$$\lambda_1 = \frac{3}{2}, \quad \lambda_2 = 0$$

由图形的对称性, $\lambda_3 = -\frac{1}{2}$, $\lambda_4 = 1$ 也合于所求.



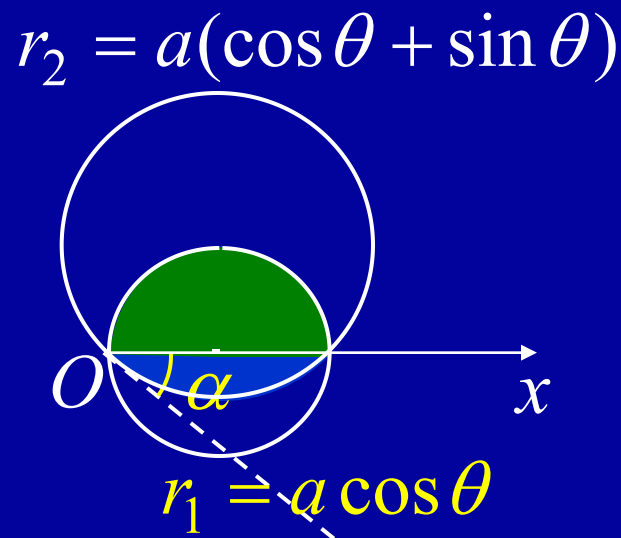
3. 求曲线 $r_1 = a \cos \theta$ 与 $r_2 = a(\cos \theta + \sin \theta)$ 所围成图形的公共部分的面积.

解: 令 $r_2(\theta) = 0$, 得 $\alpha = -\frac{\pi}{4}$

所围区域的面积为

$$\begin{aligned} S &= \frac{1}{2} \int_{-\frac{\pi}{4}}^0 [r_2(\theta)]^2 d\theta + \frac{1}{2} \cdot \pi \cdot \left(\frac{a}{2}\right)^2 \\ &= \frac{a^2}{2} \int_{-\frac{\pi}{4}}^0 (\cos \theta + \sin \theta)^2 d\theta + \frac{\pi}{8} a^2 \end{aligned}$$

$$= \frac{a^2}{2} \left(\theta - \frac{\cos 2\theta}{2} \right) \bigg|_{-\frac{\pi}{4}}^0 + \frac{\pi}{8} a^2 = \frac{a^2(\pi-1)}{4}$$

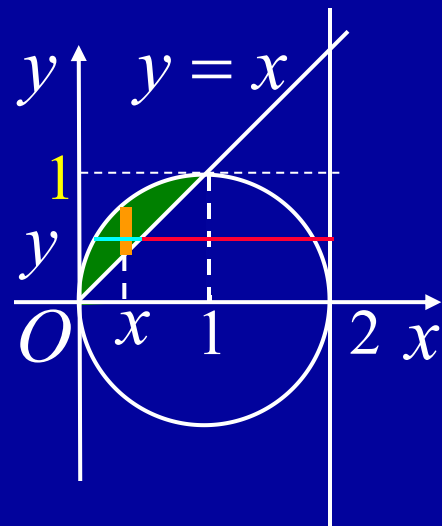


4. 设平面图形 A 由 $x^2 + y^2 \leq 2x$ 与 $y \geq x$ 所确定, 求图形 A 绕直线 $x = 2$ 旋转一周所得旋转体的体积.

提示: 选 x 为积分变量.

旋转体的体积为

$$\begin{aligned} V &= 2\pi \int_0^1 (2-x)(\sqrt{2x-x^2} - x) dx \\ &= \frac{1}{2}\pi^2 - \frac{2}{3}\pi \end{aligned}$$



若选 y 为积分变量, 则

$$V = \pi \int_0^1 [2 - (1 - \sqrt{1 - y^2})]^2 dy - \pi \int_0^1 (2 - y)^2 dy$$

